Parag Sarvoday Sahu

Research Internship in 3D Vision, Neural Rendering & 3D Understanding

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- MSc student at École Polytechnique working on 3D vision, inverse rendering, Gaussian splatting and world models.
- First-author manuscript in preparation, Pacific Graphics 2025 paper, 2nd place SIGGRAPH Asia 2025 3DGS Challenge.
- Core skills: differentiable rendering and PyTorch research code; available 22 March to 17 September 2027.

Education

École Polytechnique (IP Paris), Palaiseau, France Sep 2026 – Present
MSc&T in Visual and Creative Artificial Intelligence French Government Scholar (Charpak)
Courses: Image Analysis & Computer Vision, Digital Representation & Analysis of Shapes, Computer Animation, Foundations of ML

Indian Institute of Technology (IIT) Gandhinagar, India 2022 – 2026
B.Tech in Electrical Engineering, Minor in Computer Science CPI **9.04**/10, Dean's List
Courses: Computer Vision, Machine Learning, Matrix Methods for ML, Probability & Statistics

Publications

TensoIS: A Step Towards Feed-Forward Tensorial Inverse Subsurface Scattering for Perlin Distributed Heterogeneous Media
A. Tiwari, S. Bhardwaj, Y. Bachwana, **P. S. Sahu**, T. M. F. Ali, B. Chintalapati, S. Raman *Pacific Graphics 2025 (CGF)*

Manuscript on Multi-View Photometric Stereo with 2D Gaussian Splatting
P. S. Sahu, A. Tiwari, V. Shah, S. Raman *First author, in preparation*

Research Experience

IndiaAI Research Fellow (National Merit Fellowship) Jan 2026 – Present
IIT Gandhinagar | Inverse Rendering with Gaussian Primitives Advisor: Prof. Shanmuganathan Raman
- Formulated a novel 2D Gaussian Splatting approach to uncalibrated Multi-View Photometric Stereo (MVPS), jointly recovering geometry and materials without known lighting directions or intensities.
- Recovered surface normals at **6.09°** mean angular error on DiLiGenT-MV, and handled captures where the camera moves between light changes, unlike standard MVPS setups.
- Cut light captures **10**× (10 instead of 96 per view); under 30 minutes per object, against about 12 hours for PS-NeRF.

Research Intern, Jodhpur, India Jul 2025 – Dec 2025
3DVisLab, IIT Jodhpur Advisor: Prof. Avinash Sharma
- Explored learning-based 3D reflective-symmetry detection as a basis for symmetry-aware mesh compression.
- Applied mesh processing and deep learning to high-fidelity 3D human head reconstruction.

Projects & Competitions

Scaling Analysis of Driving World Models Oct 2026 – Present
Course research project, École Polytechnique Supervisor: Prof. Xi Wang
- Evaluating whether autoregressive driving world models gain geometric and semantic consistency with scale, across 9 VaViM checkpoints (VQ tokenizer + GPT, 185M–1.2B) on nuScenes validation scenes.
- Early results suggest the tokenizer, not model size, limits semantic fidelity.

SIGGRAPH Asia 2025 3DGS Challenge Oct 2025 – Nov 2025
Real-time 3D Gaussian Splatting for novel view synthesis paragsarvoday/DeadlineDinosaur
- Ranked **2nd of about 50 teams worldwide** with **28.04 dB** mean PSNR over 30 test scenes, using a coarse-to-fine progressive-resolution schedule that trains each scene in under 60 seconds.
- Won a **$2,000 USD** prize and presented our method in person at the SIGGRAPH Asia 2025 3DGS Workshop in Hong Kong, covering our backbone choice and hyperparameter study.

Vision Transformer Inference Engine from Scratch Mar 2026 – Apr 2026
Pure C++ / Eigen, deployed to the browser with WebAssembly paragsarvoday/ViT-Inference
- Implemented ViT inference without a deep-learning framework, with an on-device web demo.

Skills

Programming: Python, C++, WebAssembly, MATLAB
Libraries & Tools: PyTorch, gsplat, Mitsuba 3, Eigen, OpenCV, Git/GitHub, LaTeX

Honors and Awards

- **Charpak Master's Scholarship:** Selective French government award for graduate studies (1 of 19 nationwide). 2026
- **Bipin & Rekha Shah & Prof. DV Pai Scholarships:** Awarded for academic excellence at IIT. 2023, 2024

Interests

Badminton · Reading (public bookshelf↗)